

AZTHREUS SYSTEMS: PROJECT MANAGEMENT & FINANCE

Document ID: AZ-CONSOLIDATED-05 | **Classification:** Internal/Confidential **Owners:** COO, Finance & Stewardship Officer | **Review Cycle:** Annual **Jurisdictions:** Republic of the Philippines, United States of America

1. PROJECT MANAGEMENT FRAMEWORK

(AZ-PM-001 & AZ-ENG-PM-001)

1.1 The Azthreus Project Framework

Standard agile frameworks were designed for low-stakes products with short iterations. Azthreus Systems operates where a defect can cause mission loss or loss of life, and projects may span decades. The hybrid framework combines: agile iterative delivery + transparency, aerospace gate reviews + traceability (NASA Systems Engineering Handbook SP-2016-6105), and formal baselining + change control (Configuration Management, Volume 2 Part 15). Result: short, reviewable iterations with safety-critical rigor.

1.2 Project Classification

Class	Description	Governance	Examples
A: Mission-Critical	Failure = loss of life/mission	Executive Board/CTO sign-off, full gate reviews (PDR/CDR/TRR/FRR)	RhAOS, DTN flight stack, command uplink
B: Mission-Support	Failure = degraded operations	COO/Engineering Lead sign-off, PDR/CDR/TRR	Ground station tools, telemetry dashboards
C: Infrastructure	Failure = internal disruption	Simplified CCB, optional plans	CI/CD pipeline, developer tooling
D: Research	No committed production deliverable	Flexible iteration, minimal documentation	Quantum algorithm research

1.3 Project Initiation & Charter

Every project requires an approved Charter before engineering begins. Class A/B: Executive Board or CTO sign-off. Class C/D: COO or Engineering Lead. Charter contents: Project Name/ID, Category, Problem Statement, Objectives (measurable), Scope (in/out), Constraints (budget/timeline/tech/regulatory), Stakeholders, Initial Risk Assessment, Dependencies, Success Criteria.

Required Planning Documents (after Charter, before first Sprint): | Document | Required For | Owner | |-----|-----|-----| | Software Development Plan | A, B | Engineering Lead | | Configuration Management Plan | A, B | Systems Architect | | Risk Management Plan | A, B, C | Project Manager | | Quality Assurance Plan | A, B | Quality Manager | | V&V Plan | A (mandatory), B (recommended) | IV&V Lead | | Communications Plan | All Classes | Project Manager |

Documents need not be large — a Class C Software Development Plan may be two pages. The requirement is decisions made and recorded before work begins.

1.4 Sprint Structure

Cadence: 2 weeks standard. Class A with HW-in-loop: 4 weeks. Class D: flexible per Charter.

Event	Duration	Purpose
Sprint Planning	2h	Select goals, decompose backlog into tasks
Daily Standup	15 min	Progress sync, surface blockers
Sprint Review	1h	Demo completed work to stakeholders
Sprint Retrospective	1h	Inspect process, identify improvements

Sprint events are not optional. If key participants unavailable, reschedule within same day, not skip.

Definition of Ready: Requirement clearly written and understood, acceptance criteria defined and testable, dependencies identified, estimated, design decisions documented.

Definition of Done: Code written and peer-reviewed (approved MR). All unit tests pass. Code coverage requirements met (Procedure 16.1.1/16.1.2). Static analysis passes (no unresolved findings). Documentation updated. Class A: IV&V complete or formally scheduled. Demonstrated in Sprint Review and accepted by Engineering Lead.

1.5 Formal Gate Reviews (Class A/B)

Gate	Milestone	Decision
PDR	Architecture/high-level design complete	Approve detailed design
CDR	Detailed design complete	Approve development
TRR	Code complete, test environment ready	Approve formal testing
FRR	Class A only: all testing complete, ready for operational deployment	Approve deployment

Review Board: Project Sponsor, independent Systems Architect (not primary designer), Quality Manager. Class A: independent IV&V rep. PM and Engineering Lead present but do not vote.

Outcomes: Pass / Conditional Pass (action items with dates, verified by Quality Manager) / Fail (issues must be resolved, gate re-reviewed). Outcomes + docs stored as controlled documents.

[EXPANDED]

1.5.1 Gate Review Exit Criteria — Literal Checklists

Every gate review must verify that **all** of the following exit criteria are satisfied before a Pass or Conditional Pass decision. The Quality Manager verifies each criterion and records the verification outcome (Pass/Fail/Conditional) in the Gate Review Record. Any Fail criterion blocks gate passage unless the Review Board unanimously grants a Conditional Pass with a documented remediation plan and re-verification date.

A. Preliminary Design Review (PDR) Exit Criteria

#	Exit Criterion	Verification Method	Required For
PDR-01	Project Charter approved and signed by sponsor	Review signed document	Class A, B
PDR-02	Problem statement, objectives, scope, constraints documented and approved	Document review	Class A, B
PDR-03	System architecture defined at high level (block diagrams, component interactions)	Architecture review	Class A, B
PDR-04	All major interfaces identified (internal and external, including hardware interfaces for HW-in-loop)	Interface catalog review	Class A, B
PDR-05	Top-level requirements derived from mission/system requirements and traced to system-level requirements	Requirements traceability matrix review	Class A
PDR-06	Initial risk assessment completed (per §1.6) with all risks scored and assigned	Risk Register review	Class A, B
PDR-07	Technology choices documented with rationale (ADRs for major selections)	ADR review	Class A, B
PDR-08	Preliminary schedule with milestones, dependencies, and critical path	Schedule review	Class A, B
PDR-09	Resource plan (personnel, equipment, facilities, HSM/clean room if needed)	Resource allocation review	Class A, B
PDR-10	Data governance plan for training/operational data (if AI/ML system) per AZ-AI-001 §2.1	AI data governance review	Class A, B (if AI)
PDR-11	Safety requirements identified and traced to system-level safety requirements (Class A only)	Safety requirements matrix review	Class A
PDR-12	Export control classification determined for all deliverables	Legal/Compliance review	Class A
PDR-13	Open source dependency landscape assessed; no Prohibited licenses identified in planned dependencies	SBOM draft review per AZ-LEGAL-001 §6.4	Class A, B

B. Critical Design Review (CDR) Exit Criteria

#	Exit Criterion	Verification Method	Required For
CDR-01	All high-level requirements decomposed into detailed design requirements	Requirements traceability matrix (full)	Class A, B
CDR-02	Detailed design documents complete for all modules/components	Design document review	Class A, B
CDR-03	All Interface Control Documents (ICDs) signed off by responsible parties (internal) and counter-parts (external)	Signed ICDs in controlled document system	Class A, B
CDR-04	100% of interfaces mapped in the Architecture Decision Record (ADR) — no untracked interfaces	Interface-to-ADR cross-reference audit	Class A, B
CDR-05	Memory budget <80% utilization — flight software must demonstrate reserved headroom for FDIR, telemetry buffers, and degradation modes	Memory budget analysis (static analysis + worst-case runtime estimation)	Class A
CDR-06	CPU/throughput budget <70% utilization — flight software must demonstrate headroom for contingency processing	Throughput analysis (worst-case execution time profiling)	Class A
CDR-07	State machine design complete with all states, transitions, and invalid-transition handling documented	State machine table + FDIR review	Class A, B
CDR-08	Error handling strategy defined: all error codes enumerated per module, error propagation paths documented, graceful degradation verified for all failure modes	Error handling matrix review	Class A
CDR-09	Security design review complete: authentication, authorization, encryption, key management per AZ-CONSOLIDATED-02	Security design review sign-off	Class A, B
CDR-10	Data classification confirmed for all data elements per AZ-DATA-001 (L1-L4)	Data classification matrix review	Class A, B
CDR-11	Explainability approach defined and approved for AI/ML systems (T1/T2) per AZ-AI-001 §3	Explainability design review	Class A, B (if AI T1/T2)

#	Exit Criterion	Verification Method	Required For
CDR-12	Human override mechanism designed and validated for safety-critical AI per AZ-AI-001 §4	Override design review	Class A (if AI T1/T2)
CDR-13	Test strategy defined: unit, integration, system, acceptance test plans with coverage targets	Test plan review	Class A, B
CDR-14	V&V plan finalized (Class A mandatory; Class B recommended)	V&V plan review	Class A (mandatory), B (rec)
CDR-15	Configuration management plan finalized	CM plan review	Class A, B
CDR-16	SBOM (draft) includes all planned dependencies with verified SPDX license IDs	SBOM review per AZ-LEGAL-001 §6.4	Class A, B
CDR-17	AI model architecture approved by EARB (T1/T2) per AZ-AI-001 §6	EARB approval record	Class A (if AI T1/T2)

C. Test Readiness Review (TRR) Exit Criteria

#	Exit Criterion	Verification Method	Required For
TRR-01	All code committed to version control, peer-reviewed, and merged	Git log + MR review records	Class A, B
TRR-02	100% of planned unit tests passing	CI pipeline test results	Class A, B
TRR-03	Code coverage meets minimum threshold (Procedure 16.1.1.1/16.1.1.2): 80% line coverage for flight software, 70% for ground station	Coverage report review	Class A, B
TRR-04	Static analysis: zero High/Critical findings	Static analysis report review (PC-lint/Coverity/Clippy/SonarQube)	Class A, B
TRR-05	Integration test environment provisioned and verified — matching target hardware or validated simulator	Environment verification checklist	Class A
TRR-06	Hardware-in-the-loop testbed operational (if applicable) — flight hardware or emulator available, network configured, telemetry capture validated	HW-in-loop readiness checklist	Class A (if HW)
TRR-07	Test data sets prepared: normal, boundary, adversarial, out-of-distribution (T1/T2 AI)	Test data review	Class A, B
TRR-08	All planned test cases documented with expected results and pass criteria	Test case repository review	Class A, B
TRR-09	Defect tracking: all known defects triaged, none classified as ship-blocker without documented mitigation	Defect report review	Class A, B
TRR-10	V&V independence confirmed (IV&V team has no development role) — Class A	IV&V team roster verification	Class A

#	Exit Criterion	Verification Method	Required For
TRR-11	Security scan complete: no unresolved High/Critical CVEs in dependencies per AZ-CONSOLIDATED-02	CVE scan report review	Class A, B
TRR-12	AI bias audit complete for T1/T2/T3 per AZ-AI-001 §2.5 and §8	Bias audit report review	Class A, B (if AI)
TRR-13	Rollback plan documented and tested for T1/T2 AI deployments per AZ-AI-001 §2.6	Rollback plan review	Class A (if AI T1/T2)

D. Flight Readiness Review (FRR) Exit Criteria — Class A Only

#	Exit Criterion	Verification Method	Required For
FRR-01	All test phases complete: unit, integration, system, acceptance, performance, stress, endurance	Test completion report	Class A
FRR-02	All TRR action items resolved	Action item closure verification	Class A
FRR-03	All defects closed or formally accepted with documented risk acceptance by Project Sponsor	Defect disposition review	Class A
FRR-04	Mission simulation executed successfully — end-to-end scenario including nominal operations, failure injection, recovery	Mission simulation report	Class A
FRR-05	FDIR tested in all failure modes — single-point failures, common-cause failures, cascading failures all handled per FDIR specification	FDIR test report	Class A
FRR-06	Operational procedures documented and reviewed by Mission Operations team	Ops procedure review	Class A
FRR-07	Flight crew / Mission Controllers trained on all nominal and off-nominal procedures	Training completion records	Class A
FRR-08	SBOM finalized and attached to build artifact — no outstanding dependency updates unless formally accepted	Final SBOM review	Class A
FRR-09	Key management complete: flight keys generated per ceremony, distributed to key custodians, spacecraft loaded	Key ceremony record + distribution log	Class A
FRR-10	Configuration baseline established: all flight software, configuration data, and command tables under formal configuration control	CM baseline record	Class A
FRR-11	All ICDs signed and in controlled document system — no unsigned or draft ICDs	ICD registry audit	Class A
FRR-12	Export control compliance verified for all deliverables	Legal/Compliance sign-off	Class A

#	Exit Criterion	Verification Method	Required For
FRR-13	AI model deployed and kill-switch operational for T1/T2 AI systems — tested and verified by Mission Controllers	Kill-switch test record	Class A (if AI T1/T2)
FRR-14	Post-deployment monitoring plan documented — monitoring dashboards configured, alert thresholds set, on-call roster established	Monitoring readiness checklist	Class A
FRR-15	Lessons learned from previous missions/launches reviewed and applicable findings incorporated	Lessons learned review record	Class A
FRR-16	Independent V&V report complete with all findings addressed or formally accepted	IV&V report review	Class A

E. Conditional Pass Criteria

A **Conditional Pass** may be granted only when: 1. All critical-path exit criteria (marked “Required For Class A”) are satisfied. 2. Non-critical exit criteria have documented remediation plans with specific dates and owners. 3. The Quality Manager confirms that the conditions do not introduce unacceptable risk to mission safety. 4. The Review Board unanimously agrees to the Conditional Pass. 5. All conditions must be resolved and verified by the Quality Manager before the next gate review — unresolved conditions from a previous gate automatically block passage of the next gate. 6. Maximum time for condition resolution: 30 calendar days from gate review date. Extensions require COO approval.

1.6 Risk Management (Project Level)

Risk identification throughout lifecycle. All team members responsible. PM maintains Risk Register. Each entry: Risk ID, Description, Category (Technical/Schedule/Resource/External/Safety), Likelihood 1-5, Impact 1-5, Score (L x I, 1-25), Response Strategy, Mitigation Actions, Owner, Status.

Score	Action
1-6	Monitor, review at Sprint Retrospective
7-14	Active mitigation plan, weekly review
15-25	Immediate escalation to Sponsor + CTO, mitigation plan within 48h

Safety risks with Impact ≥ 4 escalated regardless of likelihood.

1.7 Change Management

Change Request (CR): Required for scope/schedule/budget changes. Process: Submit → PM assesses impact (scope/schedule/budget/risk) → Presented to CCB → Approved/conditionally approved/rejected → Incorporated into plan. **CCB Composition:** Class A/B → Project Sponsor, Engineering Lead, Quality Manager. Class C/D → PM + Engineering Lead. Approved scope changes affecting architecture/interfaces trigger Engineering Change Proposal under Configuration Management (Procedure 15.2.1.2) — CR and ECP run in parallel.

1.8 Project Closure

All deliverables accepted by Sponsor. Class A: FRR passed and documented. All open defects resolved or accepted with rationale. Lessons Learned session conducted (facilitated by non-PM) within 2 weeks — captures what went well/poorly, process improvements, technical findings for ADRs. Final documentation archived per OAIS compliance (Volume 1 Part 6). Knowledge Offload sessions completed (Procedure 14.3.1.1). Lessons Learned reports stored in corporate knowledge base, reviewed at start of similar future projects.

1.9 Sprint Metrics

Tracked per Sprint, reviewed at Retrospective: Sprint Velocity (story points completed), Defect Escape Rate (defects found in Sprint Review not caught in development), Test Coverage Delta (change in code coverage %), MR Cycle Time (average time open to merge), Blocker Count (items blocked >24h with reasons), Risk Register Changes (opened/closed/escalated).

2. EXPENSE & REIMBURSEMENT POLICY

(AZ-FIN-001)

2.1 Guiding Principles

Reasonableness: Expenses must be reasonable in context; economy and efficiency expected, premium options need justification. **Necessity:** Genuinely required for business purpose not accomplishable without it. **Transparency:** All expenses documented with receipts and clear business purpose. **Separation:** No personal benefit accumulation (points/miles) at company expense that influences spending toward more expensive options.

2.2 Approval Authorities

Amount (per transaction or trip total)	Approver
Up to PHP 5,000 / USD 100	Direct manager
PHP 5,001-50,000 / USD 101-1,000	Manager + Finance Officer
PHP 50,001-200,000 / USD 1,001-4,000	Department head + Finance Officer
Above PHP 200,000 / USD 4,000	COO + Finance Officer
Any involving CEO	Finance Officer + Executive Board Chair

2.3 Pre-Approval Required (before incurring)

International travel. Conference registration >PHP 10,000 / USD 200. Equipment purchases. Single meal/entertainment >PHP 5,000 / USD 100. Training/course enrollment. Trip budgets for international travel must be approved before booking. Emergency expenses without pre-approval: notify manager within 24h, submit post-approval immediately on return.

2.4 Business Travel

Booking: Through approved platform or designated coordinator (Finance Officer approval for outside). **Class of Service:** <4h flight → Economy. 4-7h → Economy (Premium Economy with Finance approval). >7h → Economy standard (Business class with COO approval). Employees may upgrade with personal funds; company reimburses only economy fare. Advance booking ≥14 days; last-minute exceeding comparable fares requires manager to document why.

Accommodation: Business-appropriate hotels. Max nightly: Metro Manila PHP 7,500, other PH PHP 4,500, US major cities USD 300, other US USD 200, international USD 250. Above limits: Finance Officer approval with documentation.

Ground Transport: Taxis/ride-hailing reimbursable. Personal vehicle mileage: PH → PHP 8.00/km, US → IRS standard mileage rate. Log starting point, destination, purpose, distance. Commuting not reimbursable.

Meals During Travel (per diem): Manila PHP 1,500/day, other PH PHP 1,000, US major cities USD 85, other US USD 65, international USD 75. Receipts required >PHP 300 / USD 10. Alcohol not reimbursable unless part of approved business entertainment.

Business Entertainment: Clear business purpose + all attendee names/organizations + receipt required. Per person >PHP 3,000 / USD 60 requires pre-approval. Alcohol capped at 20% of total bill. No reimbursement for purely social events. Employee-only events (team lunch/dinner) ≤PHP 2,500 / USD 50 per person; year-end/special ≤PHP 5,000 / USD 100.

International Travel Security: Manager + Finance pre-approval. Review security advisory (CISO brief for medium/high risk destinations). Consult Legal for ITAR/EAR-implicated destinations. Inform CISO of devices carried internationally — clean travel device may be required for high-risk destinations.

2.5 Professional Development & Training

40-hour annual CPE requirement funded within these budgets (per Policy 9.3.1): | Level | Annual Budget | |-----|-----| | Junior/Associate | PHP 25,000 / USD 500 | | Mid-level | PHP 50,000 / USD 1,000 | | Senior/Lead | PHP 75,000 / USD 1,500 | | Architect/Principal | PHP 100,000 / USD 2,000 |

Unused does not carry over. Pre-approved by manager, aligned with company technology domains (Procedure 9.3.1.1). Conference registration covered from training budget; travel follows Part 2.4 rules. Post-conference: written summary of key takeaways to team within 5 working days.

2.6 Remote Work Equipment

Category	Annual Limit Per Employee
Ergonomic accessories (chair, monitor stand, wrist rest)	PHP 15,000 / USD 300
Additional peripheral (monitor, keyboard, mouse)	PHP 15,000 / USD 300
Lighting/video equipment for remote meetings	PHP 7,500 / USD 150

Above limits: Finance Officer + manager approval. All company-purchased equipment remains company property, must be returned on separation regardless of location.

2.7 Submission & Reimbursement

Submitted through expense management system. Receipts required >PHP 300 / USD 10 (no receipt: written explanation). Business purpose for every line item — “Dinner” is not acceptable; “Dinner with [partner] to discuss DTN integration contract” is. Submit within 30 days; >60 days may be declined. Currency conversion: transaction date exchange rate from expense platform.

Timeline: Approved with complete docs → within 15 business days. Additional docs required → clock pauses. PH employees via corporate payroll → next payroll cycle after approval.

Corporate Cards: Selected employees (Finance Officer discretion). Same pre-approval/documentation/purpose requirements. Statements reconciled monthly with receipts by 5th business day. Personal charges must be reimbursed within 10 business days of statement date.

2.8 Non-Reimbursable

Commuting. Personal entertainment not connected to business. Alcohol >20% of approved entertainment bill. Traffic fines/violations. Personal care (haircuts, spa, gym). Clothing (except required PPE per Procedure 19.1.1.1). Expenses for spouse/partner/family not included in pre-approved event. Charitable donations. Any expense violating law or policy.

2.9 Compliance & Audit

Finance Officer conducts quarterly audits (≥15% sample). Fraudulent claims, personal-as-business, or repeated violations → repayment + disciplinary action (AZ-HR-006) + termination for serious/repeated fraud + legal referral for significant fraud. Suspected abuse: report to Finance Officer or Ethics Hotline (confidential, independent investigation).

3. CONSOLIDATED STANDARDS APPENDICES

3.1 Project Documentation Requirements Summary

Document	Class A	Class B	Class C	Class D
Project Charter	R	R	R	R
Software Development Plan	R	R	Opt	Opt
Configuration Management Plan	R	R	Opt	NR
Risk Management Plan	R	R	R	Opt
QA Plan	R	R	Opt	NR
V&V Plan	R	Rec	NR	NR
Communications Plan	R	R	R	Opt
Gate Reviews	All 4	PDR/CDR/TRR	NR	NR
Lessons Learned Report	R	R	R	R

3.2 Static Analysis Tools (CI Pipeline)

Language	Tool	Ruleset
C	PC-lint Plus / Coverity	MISRA C:2012 mandatory
Python	Pylint + Flake8	PEP 8 + project overrides
Rust	Clippy	Default + <code>clippy::pedantic</code> for safety-critical
All	SonarQube	OWASP Top 10, CWE Top 25

3.3 Code Review Checklist

Correctness: Does what description says. Edge cases handled. Error paths handled and tested.

Standards Compliance: Naming conventions. Function length limits. Return values checked. No prohibited patterns. **Documentation:** Function header comments present/accurate. Non-obvious logic explained. README updated.

Testing: Unit tests for all new logic. Error path tests. No commented-out tests without explanation.

Security: No hardcoded credentials. Input validation at external interfaces. No unapproved dependencies.

3.4 SBOM Generation Tools

Platform	Tool	Command
Python	cyclonedx-python	<code>cyclonedx-py -r -o sbom.json</code>
Rust	cargo-sbom	<code>cargo sbom</code>
C/C++	spdx-sbom-generator	<code>./generate_spdx -p myproject</code>
Containers	syft	<code>syft docker:myimage -o spdx-json</code>